DYLAN FERREIRA

DATA ENGINEER | AZURE & MICROSOFT FABRIC | PYSPARK | SQL
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LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Performance-driven **Data Engineer** with 7+ years of IT experience, including nearly 5 years designing, optimizing, and deploying enterprise data platforms and integration pipelines. Proven track record of cutting pipeline runtimes from hours to minutes using **Microsoft Fabric, Azure Data Factory (ADF), PySpark, and Delta Lake**. Specialized in Medallion architecture (Bronze/Silver/Gold), automated schema reconciliation, and high-throughput ELT/ETL processes servicing distributed multi-campus operations.

TECHNICAL SKILLS

- **Cloud & Data Platforms:** Microsoft Fabric (Lakehouse/Warehouses), Azure Data Factory (ADF), ADLS Gen2, Delta Lake, Azure SQL, SQL Server, PostgreSQL
- **Languages & Scripting:** Python, PySpark, SQL / T-SQL, PowerShell, Bash
- **Data Engineering & Architecture:** Medallion Architecture (Bronze/Silver), ETL/ELT Pipeline Development, Incremental Loads, Change Data Capture (CDC), Delta Table Merges, Data Modeling (Kimball / Star Schema), Schema Drift Handling
- **Data Integration & Tools:** REST APIs, JSON/Parquet, SQLAlchemy, pyodbc, Azure Key Vault, Power BI, Git / Version Control, Pecan AI

PROFESSIONAL EXPERIENCE

Database Administrator & Data Engineer
San Joaquin Valley College | Jan 2022 – Present (Remote)

- **Accelerated pipeline processing speed by 95%** by migrating 30+ CRM vendor ingestion routines from legacy formats to optimized PySpark in Microsoft Fabric, reducing runtimes from 2+ hours to 5 minutes.
- **Architected end-to-end Medallion (Bronze/Silver) pipelines** in Azure Data Factory and Microsoft Fabric, processing hundreds of thousands of records per execution across 26 campus operations.

- **Engineered robust Lakehouse ingestion models**, eliminating downstream Power BI reporting disruptions by actively managing Delta Lake schema drift, concurrent merge conflicts, and datatype inconsistencies.
- **Decoupled analytics workloads from core OLTP systems** by orchestrating enterprise data migrations from transactional SQL Server databases into ADLS Gen2 and Microsoft Fabric Lakehouse.
- **Automated incremental loads and historical data reconciliations**, standardizing overlapping month-to-date (MTD) and year-to-date (YTD) datasets to ensure single-source-of-truth accuracy.
- **Optimized heavy analytic and ETL queries**, tuning execution plans, indexes, joins, and partition schemes to reduce execution times from 60+ seconds to sub-second responses.
- **Deployed predictive ML pipelines** by building and operationalizing 4 student-retention and financial-risk models with Pecan AI, delivering automated scores directly to operational intervention teams.

Help Desk Specialist, Tier 1
San Joaquin Valley College | Sep 2019 – Dec 2021 (Visalia, Ca / Hybrid)

- Managed system provisioning, role-based access control, and account governance across Active Directory and Azure AD for 26 campuses.
- Automated administrative provisioning workflows using PowerShell, authoring comprehensive technical documentation adopted department-wide.

IT & Systems Support Roles
IT Field Technician (MTI) & Systems Repair Technician | Jan 2018 – Sep 2019

- Configured, deployed, and supported hardware infrastructure, network connectivity, and operating systems across commercial client sites.

KEY PROJECTS

- **Canvas LMS Data 2 Pipeline (Fabric / Python / REST):** Designed a daily automated pipeline in Fabric synchronizing 49 LMS relational tables via Python and REST APIs. Built automated reconciliation scripts to validate row-counts and ensure data completeness against upstream APIs.
- **Automated Enterprise Data Ingestion (Azure Key Vault / Document Intelligence):** Built secure ingestion micro-pipelines using Python, T-SQL, and Azure Document Intelligence to parse unstructured document data into structured tabular datasets, centralizing credential management in Azure Key Vault.
- **Automated Pipeline Data Reconciler (Fabric / Power BI):** Constructed an automated regression validation tool comparing modern Fabric pipeline outputs

against legacy manual flat-file imports, catching discrepancies prior to production cutover.

EDUCATION & CERTIFICATIONS

- **Associate of Science in Information Technology** — San Joaquin Valley College (2017 – 2018)
- **Microsoft Certified:** Azure Data Fundamentals (DP-900)
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